

MA 563: Statistical Computation and Simulation

Course Code: MA 563

Course Name: Statistical Computation and Simulation

L-T-P-C: 3-0-2-4

Prerequisites: Probability and Statistics (MA 524) and/or IC 252

Intended for: B.Tech 3rd and 4th year/ MS/MSc/PhD

Mutual Exclusion: NA

Elective or Core: Elective

Approval: 61st BoA

Preamble

Statistical computation and simulation play a key role in modern statistical analysis, data science, and machine learning, particularly when analytical solutions are unavailable or impractical. This course is designed to introduce students to fundamental computational tools in statistics, with a strong emphasis on simulation-based methods and optimization techniques that are routinely used in statistical modeling and inference. The course balances theoretical foundations with practical implementation, enabling students to understand both why computational methods work and how to apply them effectively. Starting from random number generation and simulation of random variables, the course progressively introduces Monte Carlo methods, resampling methods, and algorithmic approaches such as EM. Special emphasis is placed on Bayesian computational approaches, including Monte Carlo methods and Markov chain Monte Carlo (MCMC) techniques, which are essential for modern Bayesian inference. Overall, the course equips students with a unified computational perspective on statistical inference, optimization, and simulation, preparing them for advanced research and practical applications in statistics, machine learning, and related fields.

Course Modules

- **Random Number Generation and Simulation Foundations:** Pseudo-random number generators, uniform and non-uniform random variable generation, Inverse transform method, acceptance–rejection method, Box–Muller transformation, ratio-of-uniforms method, diagnosing randomness, and simulation error. (8 Lectures)
- **Monte Carlo Methods:** Monte Carlo estimation and integration, law of large numbers and Monte Carlo error, crude Monte Carlo, importance sampling (simple & weighted), variance reduction ideas (basic level), and practical Monte Carlo diagnostics. (8 Lectures)
- **Resampling and Simulation-Based Statistical Inference:** Bootstrap principles, parametric vs nonparametric bootstrap, Cross-validation, Jackknife (brief), Simulation-based uncertainty quantification, Computational trade-offs, likelihood-based inference via simulation, Parametric simulation for inference, Confidence intervals via simulation, Hypothesis testing using Monte Carlo, Permutation tests, Bootstrap-based inference (deep dive), Simulation vs asymptotic inference. (10 Lectures)
- **Iterative Computational Algorithms:** Role of simulation in iterative inference, stochastic approximation ideas, Likelihood-based computation, MM (Majorization–Minimization) algorithms, Mixture models (Gaussian mixtures), EM algorithm, Simulation-based interpretation of EM. (6 Lectures)
- **Bayesian Computation and MCMC:** Basics of Bayesian Inference, computational motivation for Bayesian inference, Monte Carlo methods in Bayesian statistics, Markov chains and stationary distri-

butions, Metropolis–Hastings algorithm, Gibbs sampling (basic structures), posterior diagnostics and convergence assessment. (10 Lectures)

Lab Class

The lab component of this course focuses on the practical application of concepts via hands-on implementation with real and simulated datasets. It aims to enhance students' understanding of concepts and strengthen their ability to use statistical simulation and computational approaches.

Project and Case Studies

This course will give a certain weight to project and case studies, applying statistical computing methods in day-to-day real-life datasets.

Text Books

1. Ross, S. M. *Simulation* (5th ed.). Academic Press (2012).
2. Robert, C. P., & Casella, G. *Monte Carlo Statistical Methods*. Springer (2004).

Reference

1. James, G., Witten, D., Hastie, T., & Tibshirani, R. *An Introduction to Statistical Learning*. Springer (2013).
2. Efron, B., & Tibshirani, R. J. *An Introduction to the Bootstrap*. CRC Press (1994).